## 1. Part 1: Fundamentals

### Topics Covered

- Variables and Constants
- Input and Output
- Data Types
- Operators
- Type- Casting

### Part A – Basic Problems

1. Write a C++ program to display "Hello, World!" on the screen.
2. Write a program to declare an integer variable and display its value.
3. Write a program to read an integer from the user and print it.
4. Write a program to read two integers and display their sum.
5. Write a program to calculate the square of a given number.
6. Write a program to calculate the area of a rectangle using length and breadth.
7. Write a program to convert temperature from Celsius to Fahrenheit
$$F = \left(\frac{9}{5} \times C\right) + 32$$
8. Write a program to calculate the simple interest using principal, rate, and time, using the formula:
$$\text{Simple Interest (SI)} = \frac{P \times R \times T}{100}$$
9. Write a program to display the size of basic data types (`int`, `float`, `double`, `char`).
10. Write a program to perform addition, subtraction, multiplication, and division of two numbers.
11. Write a program to find the remainder of two integers using the modulus operator.

### Part B – Moderate Problems (Conceptual)

1. Write a program to swap two numbers using a third variable.
2. Write a program to swap two numbers without using a third variable.
3. Write a program to calculate the average of three numbers entered by the user.
4. Write a program to calculate gross salary using basic salary and allowances.
$$\text{Gross Salary} = B + \text{HRA} + \text{DA} + \text{Other Allowances}$$
5. Write a program to evaluate the expression:
$$(a + b)^2 = a^2 + 2ab + b^2$$
6. Write a program to demonstrate **implicit type casting** by adding an integer and a float value and displaying the result.
7. Write a program to demonstrate **explicit type casting** by converting a float value into an integer and displaying both values.
8. Write a program to convert a character into its corresponding **ASCII value** using type casting.
9. Write a program to calculate the average of two integers and display the result as a **floating-point value** using explicit type casting.
10. Write a program to check whether a character is a vowel or consonant.
11. Write a program to input marks of three subjects and calculate total and percentage.
12. Write a program to calculate the **area** and **circumference** of a circle using the following formulae:
$$\text{Area of Circle} = \pi r^2$$
$$\text{Circumference of Circle} = 2\pi r$$
where $r$ is the radius of the circle and $\pi = 3.14159$.
13. Write a program to find the **third angle of a triangle** using the formula:
$$\text{Third Angle} = 180^\circ - (\text{First Angle} + \text{Second Angle})$$
where the sum of the interior angles of a triangle is 180°.
14. Write a C++ program to compute the **quotient** and **remainder** of two integers.
15. Write a C++ program to evaluate the following expression using operator precedence and associativity rule and display the final result:
$$\text{Result} = ((10 + 5) \times 2 > 20) \ \&\& \ (30 - 10 == 20)$$
16. Write a C++ program to evaluate the following expression using operator precedence and associativity rule and display the final result:
$$\text{Result} = (25 \div 5 + 3 \times 2 \leq 15) \ || \ (!(8 - 3 > 10))$$

### Part C - Viva / Interview Questions

1. What is C++? How is it different from C language?
2. What is the purpose of the `main()` function in a C++ program?
3. Explain the difference between a variable and a constant.
4. What are the rules for naming variables in C++?
5. Explain the use of `cin` and `cout`.
6. What is a data type? Name basic data types in C++.
7. What is the difference between `int`, `float`, and `double`?
8. What does the `sizeof` operator do?
9. Explain arithmetic operators with examples.
10. What is the difference between `/` and `%` operators?
11. What is integer division in C++?
12. Explain relational operators. Name any four relational operators.
13. What are logical operators? Explain `&&`, `||`, and `!`.
14. What will be the output of a logical expression in C++?
15. What is operator precedence? Why is it important?
16. Explain BODMAS rule with an example.
17. What is associativity of operators?
18. What happens if parentheses are removed from an expression?
19. What is type casting?
20. Differentiate between implicit and explicit type casting.
21. Give an example where type casting is necessary.
22. How is a character converted to its ASCII value in C++?
23. Why is explicit type casting needed while calculating average of integers?
24. What is modulus operator? Where is it used?
25. What will happen if divisor is zero during division?
26. What is the output type of relational expressions?
27. Explain the difference between compile-time error and runtime error.
28. What are tokens in C++?
29. Why are comments used in programs?
30. What header file is required for input-output operations?
31. What is the use of semicolon in C++?

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